Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

LLM evaluation engineer who built AgentProbe - an **open-source automated eval pipeline** with LLM-as-judge scoring, regression tracking, CI/CD integration, and structured 5-metric output evaluation. The same system Grafana's AI/ML team needs to build internally: automated eval runs, golden test sets, anomaly detection, and a dashboard that makes quality visible across runs. 8+ years production Python. Open-source-first. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Demo: *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

LLM Evaluation Frameworks | LLM-as-Judge Design | Automated Eval Pipelines | Regression Tracking | CI/CD Eval Integration | Structured Output Scoring | Prompt Engineering | RAG Pipeline Evaluation | LLM Observability | Python / FastAPI | Multi-Provider LLM Routing | Open-Source Development

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Built **internal debugging and performance analysis tools** adopted across multiple squads - reducing debugging time by **40%** and establishing quality gates used in ongoing sprints; same discipline as building eval tooling other teams adopt.
- Applied **Python, OpenAI API, and PyTorch** to automate HUD income limit data transformation workflows, replacing manual Excel operations across **1.6M records** - production ML/AI integration at scale.
- Improved production query performance by **30%** across 20+ projects; maintained **99.5% uptime** for 6+ months via TDD/BDD quality gates; delivered 20+ modules across **120+ Agile sprints** for 100K+ monthly active users.
- Encrypted **100% of resident PHI** (phone numbers, SSNs) via Libsodium across the full database; migrated thousands of client documents to AWS S3 resolving **90% of legacy storage constraints**.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Translated technical concepts for 30+ students; developed supplementary tooling and exercises adopted across the course - the same cross-team enablement the role requires.

PROJECTS

AgentProbe: Automated LLM Evaluation & Observability Platform OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · Anthropic · OpenAI · Groq · Ollama · Docker · GitHub Actions

Production eval framework built specifically around the problems this role solves. **LLM-as-judge engine**: 5-metric structured scoring (factual accuracy, hallucination, retrieval quality, completeness, compliance) - calibrated judge prompts, reproducible across runs. **Regression tracking**: anomaly flags surface quality drops before they compound; trend dashboard tracks scores across all runs. **Golden test sets**: fixed eval benchmarks run on every build for consistent regression detection. **CI/CD integration**: GitHub Actions gates block deploys when eval scores breach threshold. **Context-window management**: multi-provider routing (Anthropic, OpenAI, Groq, Ollama) with dynamic selection. Open-source, fully documented, designed for teams to adopt. *github.com/venkatalakshmiparimala/AgentProbe*

CrashClarity: Predictive ML Pipeline - Statistical Evaluation on 100K+ Records

Python · R · Random Forest · Logistic Regression · Feature Engineering · Permutation Importance · Statistical Analysis

Full ML lifecycle with rigorous evaluation: feature importance validation (MDI + Permutation Importance), model comparison (RF vs LR), statistical significance testing. **85%+ accuracy.** Demonstrates the evaluation discipline - not just training, but systematic measurement and reproducible comparison.

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings

NLP pipeline mapping natural language queries to FHIR APIs via LLMs and BERT embeddings. HIPAA-compliant RBAC. **~60% retrieval time reduction.** *github.com/venkatalakshmiparimala/AI_on_FHIR*

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, ML evaluation, experimental design, reproducibility, data pipelines at scale

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, R, Bash, HTML/CSS

LLM Eval & AI: LLM-as-Judge, Automated Eval Pipelines, Regression Tracking, LangChain, LangGraph, Multi-Provider LLM (Anthropic, OpenAI, Groq, Ollama), RAG Pipelines, ChromaDB, Prompt Engineering, FastAPI

ML & Modeling: scikit-learn, PyTorch, TensorFlow, Random Forest, Feature Engineering, Permutation Importance, Statistical Evaluation, NumPy, Pandas, BERT, spaCy, Hugging Face

Data & Cloud: AWS (S3, EC2, RDS, Glue), Azure, PostgreSQL, MySQL, Docker, CI/CD (GitHub Actions), RabbitMQ

Quality & Process: TDD/BDD, Integration Testing, Open-Source Development, Git, JIRA, Agile/Scrum